Drawing Cumulative Frequency Diagrams

Exam code: 0607

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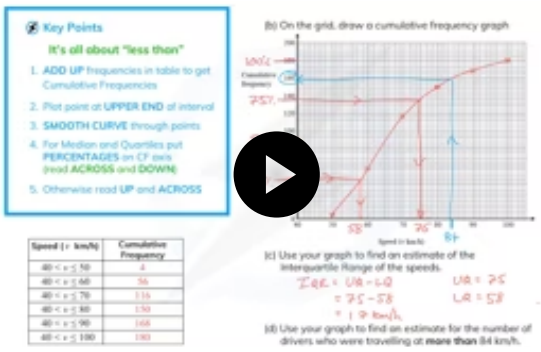

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Drawing cumulative frequency diagrams

Help me understand

What is a cumulative frequency diagram?

- A **cumulative frequency diagram** is a way of representing **grouped continuous data**
- A cumulative frequency diagram can be used to **estimate** other **statistical values**
  - For example the **median**, **quartiles** or **percentiles**

How do I draw a cumulative frequency diagram?

- This is best explained with an example
  - The times taken to complete a short general knowledge quiz taken by 50 students are shown in the table below:

Time taken (s seconds)	Frequency
$25 \leq s < 30$	3

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$35 \le s < 40$	17
$40 \le s < 45$	12
$45 \le s < 50$	7
$50 \le s < 55$	3
Total	**50**

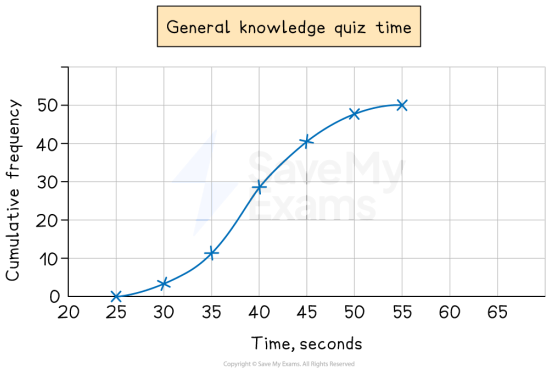
- Then the **cumulative frequency** is the running total of the frequencies

Time taken (s seconds)	Frequency	Cumulative Frequency
$25 \le s < 30$	3	3
$30 \le s < 35$	8	3 + 8 = 11
$35 \le s < 40$	17	11 + 17 = 28
$40 \le s < 45$	12	28 + 12 = 40
$45 \le s < 50$	7	40 + 7 = 47
$50 \le s < 55$	3	47 + 3 = 50
Total	**50**	

- We can now draw the **cumulative frequency diagram**
  - The most important part is that **cumulative frequency** is plotted against the **end** (**upper bound**) of the **class interval**
    - The end of the class interval is the *x*-coordinate
    - The cumulative frequency is the *y*-coordinate
    - For the above example the first two points to plot would be (30, 3) and (35, 11)
  - To explain this, consider the second row ($30 \le s < 35$)
    - The 8 students in this group could have taken **any** time between 30 and 35 seconds
    - They cannot **all** be guaranteed to have been accounted for until we reach 35 seconds
  - Once all points from the table are plotted, a point for the **start** needs to be added
    - This will be at the lowest time from the table
    - i.e. at 25 seconds with a cumulative frequency of 0
    - Plot the point (25, 0)
  - Join points up with a **smooth curve** (this takes some practice)
    - Make sure it goes through all of the marked points

back towards the x-axis

- Here is the final cumulative frequency diagram for the quiz times



Example of a cumulative frequency diagram

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